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Course: **Statistics**

1 Basic probability theory

Exercise 1.1. For each of the following experiments, describe the sample space.

(a) Toss a coin four times.

(b) Count the number of insect-damaged leaves on a plant.

(c) Measure the lifetime (in hours) of a particular brand of light bulb.

(d) Record the weights of 10-day-old rats.

(e) Observe the proportion of defectives in a shipments of electronic components.

Solution. The sample space S is the set of all possible outcomes.

(a) For each coin toss in the series we receive either Head (H) or Tail (T). The example of the realization of the series of four tosses: THHH, meaning 1st toss gives T and the remaining tosses give H. The sample space S is the set of all such combinations. The cardinality of this set is $2^4 = 16$: on each position (toss) we have either T or H (2 options) and we observe four positions.

(b) The number of leaves should be a nonnegative integer. Thus, $S = \{0, 1, 2, 3, \ldots\}$.

(c) The lifetime may be less than an hour or some integer number of hours plus the remaining fraction of an hour. Moreover, the lifetime cannot be negative. Thus, $S = (0, T]$ for some constant $T < \infty$.

(d) Consider grams. Then we may say $S = (0, \infty)$. We may also define some upper bound, for example K. Then $S = (0, K]$.

(e) Define n to be the number of items in the shipment. Since we need a proportion, the sample space is $S = \{\frac{0}{n}, \frac{1}{n}, \frac{2}{n}, \frac{3}{n}, \ldots, \frac{n}{n}\}$.

Exercise 1.2. Verify the following identities:

(a) $A \setminus B = A \setminus (A \cap B) = A \cap B^c$

(b) $B = (B \cap A) \cup (B \cap A^c)$

(c) $B \setminus A = B \cap A^c$

(d) $A \cup B = A \cup (B \cap A^c)$

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Solution. When working with set expressions, Venn diagrams might be useful. However, the illustration does not constitute a formal proof and thus is not sufficient as an answer. To prove that the expression is true, it is necessary to explicitly show both directions: if $x \in$ lhs (left-hand side) then $x \in$ rhs (right-hand side) *and* if $x \in$ rhs then $x \in$ lhs.

(a) $A \setminus B \iff x \in A$ and $x \notin B \iff x \in A$ and $x \notin A \cap B \iff x \in A \setminus (A \cap B)$. At the same time, $x \in A$ and $x \notin B \iff x \in A$ and $x \in B^c \iff x \in A \cap B^c$.

(b) By definition $A \cup A^c = S$. Then using the Distributive Law $(B \cap A) \cup (B \cap A^c) = B \cap (A \cup A^c) = B \cap S = B$.

(c) $B \setminus A \iff x \in B$ and $x \notin A \iff x \in B$ and $x \in A^c \iff x \in B \cap A^c$.

(d) $A \cup B = A \cup [(B \cap A) \cup (B \cap A^c)] = A \cup (B \cap A) \cup A \cup (B \cap A^c) = A \cup [A \cup (B \cap A^c)] = A \cup (B \cap A^c)$.

Exercise 1.3. For events A and B, find formulas for the probabilities of the following events in terms of the quantities $\mathbb{P}(A)$, $\mathbb{P}(B)$, and $\mathbb{P}(A \cap B)$:

(a) either A or B or both,

(b) either A or B but not both,

(c) at least one of A or B,

(d) at most one of A or B.

Solution.

(a) "either A or B or both" means $A \cup B$, so we have

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

This is because $A \cup B = A \cup (B \cap A^c)$ (see previous exercise), from it $\mathbb{P}(A \cup B) = \mathbb{P}(A \cup (B \cap A^c)) = \mathbb{P}(A) + \mathbb{P}(B \cap A^c) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$. In the last expression, the second equality uses the fact that A and $B \cap A^c$ are disjoint; the last equality comes from rearranging $\mathbb{P}(B) = \mathbb{P}((B \cap A) \cup (B \cap A^c)) = \mathbb{P}(B \cap A) + \mathbb{P}(B \cap A^c)$. It is easier to draw the diagram and note the fact that we count the area of the intersection two times if we do not subtract $\mathbb{P}(A \cap B)$.

(b) "either A or B but not both" is $(A \cap B^c) \cup (B \cap A^c)$, so we have

$$\begin{aligned}\mathbb{P}((A \cap B^c) \cup (B \cap A^c)) &= \mathbb{P}(A \cap B^c) + \mathbb{P}(B \cap A^c) \\ &= [\mathbb{P}(A) - \mathbb{P}(A \cap B)] + [\mathbb{P}(B) - \mathbb{P}(B \cap A)] \\ &= \mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(A \cap B).\end{aligned}$$

The easy way is to draw a diagram and note that this probability is equal to $\mathbb{P}(A \cup B) - \mathbb{P}(A \cap B)$.

(c) "at least one of A or B" is $A \cup B$, so we have

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

(d) "at most one of A or B" is $(A \cap B)^c$, so we have

$$\mathbb{P}((A \cap B)^c) = 1 - \mathbb{P}(A \cap B).$$

Note that this event also includes the possibility of not A and not B.

Exercise 1.4. Consider two different setups:

(a) A fair dice is cast until a 6 appears. What is the probability that it must be cast more than five times?

(b) A pair of events A and B cannot be simultaneously mutually exclusive and independent. Prove that if $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$, then:

- if A and B are mutually exclusive, they cannot be independent,
- if A and B are independent, they cannot be mutually exclusive.

Solution.

(a) It must be cast more than five times if we do not observe the appearance of 6 on first five casts. We do not observe 6 on each cast with probability 5/6. Since casts of a die are independent the probability to not receive 6 in first five casts is $(5/6)^5 \approx 0.4$.

(b)
- Let A and B be mutually exclusive. Suppose, for the sake of contradiction, that A and B are independent, then $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. However, from the initial property $A \cap B = \emptyset$ and $\mathbb{P}(A \cap B) = 0$ (mutually exclusive). Since we are given events A and B with positive probabilities, we come to a contradiction $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) = 0$ with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$. Thus, mutually exclusive events cannot be independent.
- Independence of A and B together with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$ gives $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) > 0$. It means that $A \cap B \neq \emptyset$. Therefore, A and B are not mutually exclusive by definition. Thus, independent events cannot be mutually exclusive.

Exercise 1.5. Two coins, one with $\mathbb{P}(\text{head}) = u$ and one with $\mathbb{P}(\text{head}) = w$, are to be tossed together independently. Let

$$p_0 = \mathbb{P}(0 \text{ heads occur}), \quad p_1 = \mathbb{P}(1 \text{ heads occur}), \quad p_2 = \mathbb{P}(2 \text{ heads occur}).$$

Can u and w be chosen such that $p_0 = p_1 = p_2$? Prove your answer.

Solution. Start with probabilities: $p_0 = (1-u)(1-w)$, that is, first is T and second is T; $p_1 = (1-u)w + u(1-w)$, that is, first is T and second is H or first is H and second is T; and $p_2 = uw$, that is, both are H. Equating these probabilities, we receive

$$p_0 = p_2 \Rightarrow u + w = 1, \quad p_1 = p_2 \Rightarrow uw = \frac{w+u}{3}.$$

These together imply

$$u(1-u) = \frac{1}{3}.$$

This equation has no solution in the real numbers (more specifically, the right-hand side must be $\leq \frac{1}{4}$ for the solution in real numbers to exist). Thus, we cannot choose legitimate u and w to satisfy the conditions.